

# Secure Coding Review

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**DOMAIN:** Cyber Security

## INTRODUCTION

Secure Coding Review is the process of examining source code to identify security vulnerabilities, coding errors, and weaknesses that could be exploited by attackers. It helps developers build reliable and secure software by following secure coding practices.

## OBJECTIVE

- Identify security vulnerabilities in code.
- Detect insecure coding practices.
- Improve application security.
- Ensure compliance with coding standards.

## TOOLS USED

- Python 3
- IDLE Python
- Windows Operating System
- SQL Query Demonstration

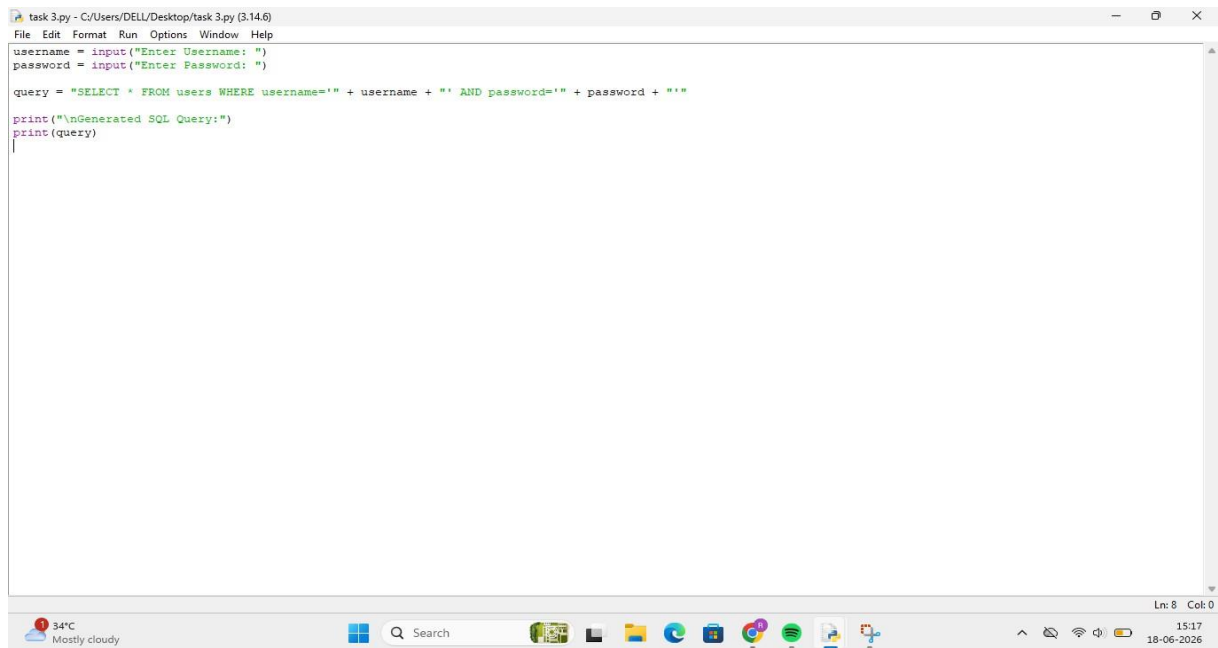
## PROGRAM CODE

```
username = input("Enter Username: ")  
password = input("Enter Password: ")
```

```
query = "SELECT * FROM users WHERE username='" + username + "' AND password='" +  
password + "'"
```

```
print(query)
```

# OUTPUT SCREENSHOTS



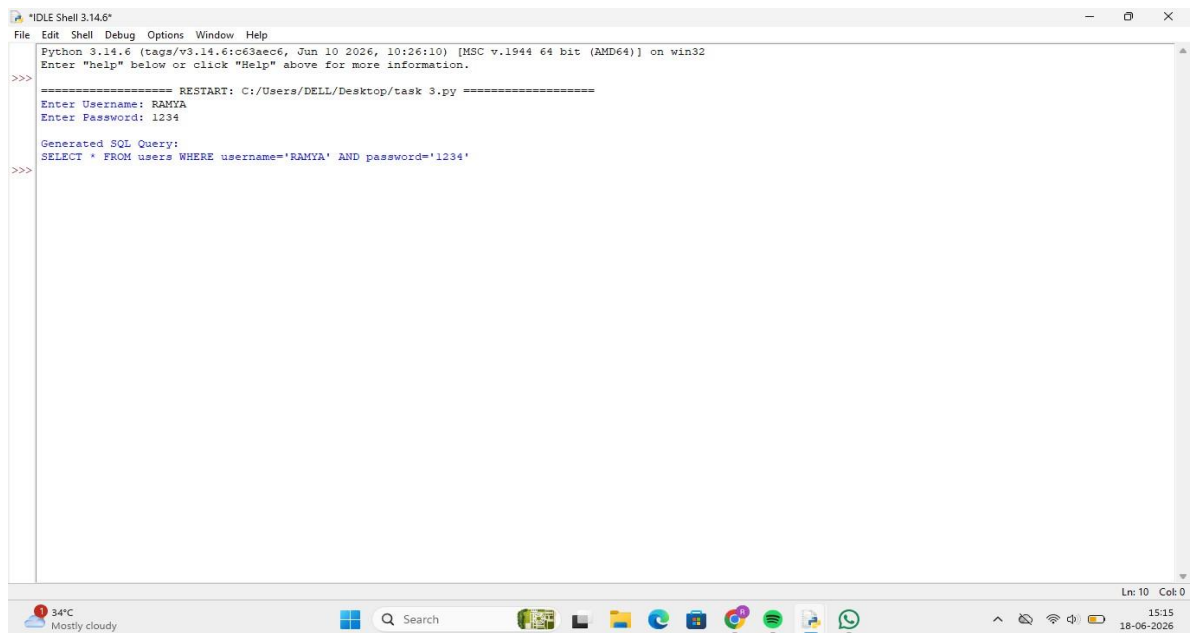
A screenshot of a text editor window titled "task 3.py - C:/Users/DELL/Desktop/task 3.py (3.14.6)". The editor contains a Python script with the following code:

```
File Edit Format Run Options Window Help
username = input("Enter Username: ")
password = input("Enter Password: ")

query = "SELECT * FROM users WHERE username='" + username + "' AND password='" + password + "'"

print("\nGenerated SQL Query:")
print(query)
```

The Windows taskbar at the bottom shows the system tray with a temperature of 34°C, weather "Mostly cloudy", and the date/time "15:17 18-06-2026".



A screenshot of the IDLE Shell window titled "IDLE Shell 3.14.6". The shell shows the execution of the Python script from the previous screenshot. The output is as follows:

```
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/DELL/Desktop/task 3.py =====
Enter Username: RAMYA
Enter Password: 1234
Generated SQL Query:
SELECT * FROM users WHERE username='RAMYA' AND password='1234'
>>>
```

The Windows taskbar at the bottom shows the system tray with a temperature of 34°C, weather "Mostly cloudy", and the date/time "15:15 18-06-2026".

## **FEATURES**

- Reviews source code for vulnerabilities.
- Detects SQL Injection risks.
- Improves software security.
- Supports secure development practices.

## **ADVANTAGES**

- Enhances application security.
- Reduces security risks.
- Improves code quality.
- Helps prevent cyber attacks.

## **APPLICATIONS**

- Secure Software Development
- Cybersecurity Audits
- Vulnerability Assessment
- Code Quality Review
- Educational Projects

## **CONCLUSION**

The Secure Coding Review project demonstrates how insecure coding practices, such as directly concatenating user input into SQL queries, can introduce security vulnerabilities like SQL Injection. Reviewing and improving code using secure coding techniques helps build safer and more reliable applications.